

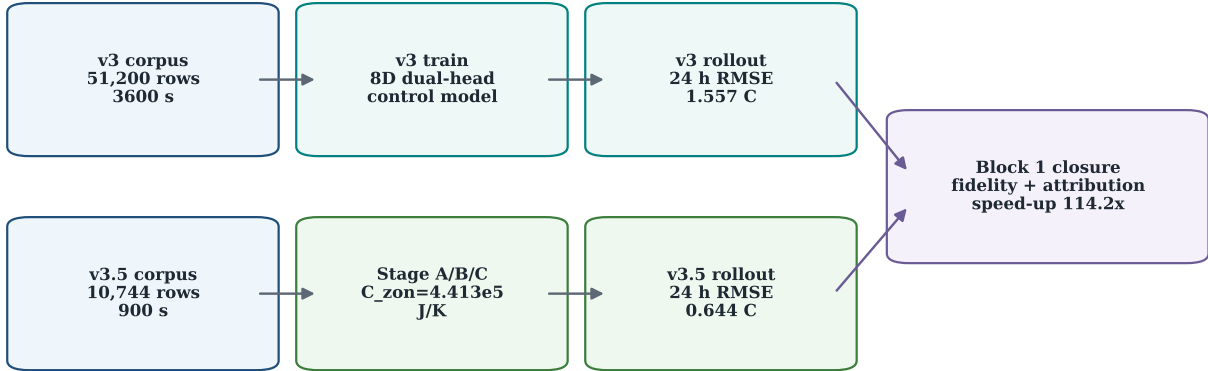
4 Digital-Twin Fidelity and the RL-Utility Paradox

This Results I section reports the Block 1 evidence chain exactly in the order used by the reproducibility roadmap. Its purpose is to establish the empirical foundation for the central claim of the article: predictive fidelity and reinforcement-learning training utility are related but not identical objectives. Block 1 proves the digital-twin side of this claim. It shows how a compact control-oriented v3 surrogate and a physically informed v3.5 surrogate differ in architecture, calibration, data resolution, rollout accuracy, and runtime feasibility. The downstream question—whether the more accurate twin is also the better PPO training environment—is intentionally carried forward to Results II, where live BOPTEST controller transfer is evaluated.

Roadmap boundary and artifact chain. Block 1 asks whether the surrogate family is credible as a fast, physically interpretable digital twin. It does not yet ask whether a controller trained on that surrogate transfers to the live BOPTEST RTE. The executed roadmap path is:

1. collect and train the v3 direct- T_{sup} surrogate using `evaluation/run_block1.py collect-data` and `v3-train`;
2. prepare the 15-minute corpus and run v3.5 Stage A/B/C inverse calibration using `prepare-15min` and `v35-calibrate -preset canonical`;
3. retrain v3 on the same 15-minute corpus using `v3-train-15min` to bound the calibration claim;
4. build the Hou-and-Evins reporting tables and speed benchmark using `build-reports` and `speed-benchmark`.

Roadmap-derived Block 1 artifact chain



Controller failure, hybrid PPO, and transfer-gap diagnostics are intentionally excluded from Results I; they start in Block 2.

Figure 1: Roadmap-derived Block 1 artifact chain. The figure reports the actual row counts, 24 h rollout errors, identified physical parameter, and speed benchmark values used to close Results I.

Table 1: Nomenclature and SI units for Block 1.

Symbol	Unit	Meaning
T_{zone}	$^{\circ}\text{C}$	zone air temperature
T_{amb}	$^{\circ}\text{C}$	ambient air temperature
T_{sup}	$^{\circ}\text{C}$	supply-air temperature command
C_{zon}	J K^{-1}	zone thermal capacitance
$\dot{Q}$	W	net zone heat flow
P	W	total HVAC electrical power
Δt	s	control / integration step
R^2	–	coefficient of determination
CV(RMSE)	%	coefficient of variation of the RMSE
NMBE	%	normalized mean bias error

4.1 Direct Supply-Temperature Control Interface

All Block 1 surrogates share a single *direct supply-temperature* (direct- T_{sup}) control parameterization: the policy commands the supply-air temperature setpoint directly, and the surrogate consumes that setpoint as a boundary condition instead of modelling the upstream valve/coil actuator chain. The two-dimensional action $a = (a_0, a_1) \in [-1, 1]^2$ maps affinely to physical commands,

$$T_{\text{sup}} = 18 + \frac{a_0+1}{2} (35 - 18) [^{\circ}\text{C}], \quad u_{\text{fan}} = \frac{a_1+1}{2} \in [0, 1], \quad (1)$$

and every surrogate exposes the same transition signature

$$f : (T_{\text{zone}}, T_{\text{amb}}, h, d, a_0, a_1) \mapsto (\hat{T}_{t+1}, \hat{P}_{t+1}), \quad (2)$$

with $\hat{P} \geq 0$ enforced by a Softplus output and capped at $P_{\text{max}} = 5500$ W. A single adapter wraps four interchangeable backends behind (2)—legacy v3, raw v3.5, calibrated v3.5, and the hybrid pairing—which is what makes the like-for-like fidelity comparisons in this section well posed: only the internal dynamics change, never the control interface or the observation map.

Table 2: Modelling assumptions of the direct- T_{sup} control interface.

Assumption	Statement
Actuator abstraction	Supply-air temperature is commanded directly; valve/coil dynamics are not modelled beyond the Stage A latency term.
Command bounds	$T_{\text{sup}} \in [18, 35] ^{\circ}\text{C}$; fan command $u_{\text{fan}} \in [0, 1]$.
Power non-negativity	$\hat{P} = \text{Softplus}(\cdot) \geq 0$, capped at $P_{\text{max}} = 5500$ W.
State support	Zone and ambient temperatures normalized to $[-1, 1]$ over $[15, 35]$ and $[-10, 40] ^{\circ}\text{C}$.
Interface invariance	All four backends obey (2); only the internal dynamics differ.

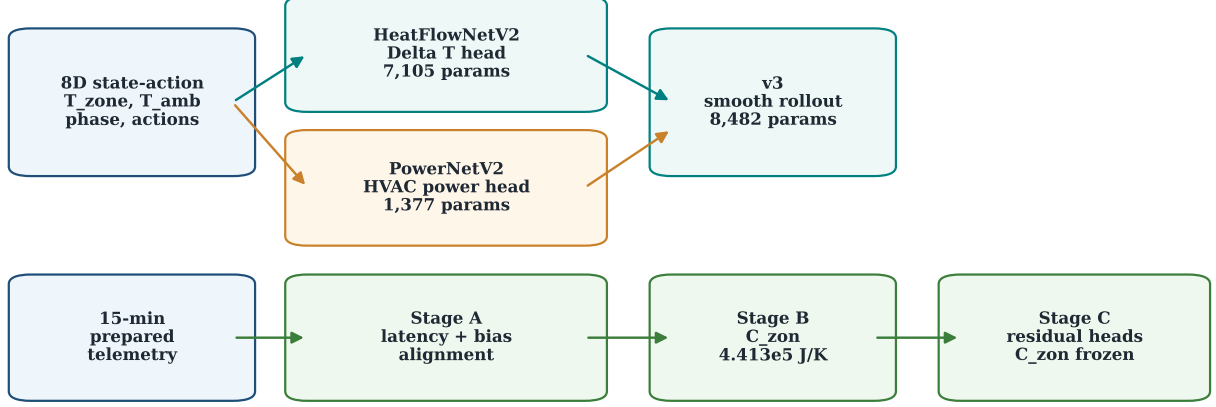
4.2 Control-Oriented Surrogate Architecture (v3)

The v3 surrogate is a compact control-oriented direct- T_{sup} model. Its role is not to be the most accurate long-horizon forecaster; its role is to provide smooth and fast rollout dynamics for downstream PPO experiments. In contrast, v3.5 is a physics-informed extension whose Stage B parameter is interpretable as the zone thermal capacitance C_{zon} .

Table 3: Block 1 data corpora referenced by the roadmap.

Dataset	Rows	Step (s)	Policy mix	Scenario mix
v3 canonical corpus	51,200	3600	cool, heat, mixed, random	autumn, spring, summer, winter
v3.5 calibration corpus	10,744	900	hdrl, thermostatic	peak_heat_window, typical_heat_window
15-min exploration corpus	48,384	900	cool, heat, mixed, pulse, random, thermostatic_noise	autumn, spring, summer, winter

Surrogate roles: v3 smooth rollout model versus v3.5 physical twin



The section compares predictive validity and physical identification; downstream controller utility is evaluated later.

Figure 2: Surrogate role separation inside Block 1. v3 is the compact dual-head rollout model; v3.5 adds Stage A/B/C physical identification and residual-head calibration.

Table 4: Architecture and calibration summary for the two Block 1 surrogate roles.

Surrogate	Role	Architecture / physics	Corpus	Training or calibration protocol
v3 direct-TSup	Fast control-oriented rollout dynamics	8D dual-head MLP: HeatFlowNetV2 (7,105 params, LayerNorm/Tanh/residual), PowerNetV2 (1,377 params, Tanh/Softplus); total 8,482 params	51,200 hourly transitions	AdamW, lr=1e-3, batch=256, epochs=500
v3.5 calibrated	Physics-informed predictive twin	RC-NeuralODE backbone with positive C_zon reparameterization and residual temperature/power heads	10,744 prepared 15-min transitions	Stage A latency/bias; Stage B 120 epochs; Stage C 60+80 epochs; C_zon=4.413e5 J/K

The v3 transition model is

$$\Delta\hat{T}_{t+1} = f_{\theta,T}(x_t), \quad \hat{P}_{t+1} = f_{\theta,P}(x_t), \quad \hat{T}_{t+1} = T_t + \Delta\hat{T}_{t+1}. \quad (3)$$

It is trained by a multi-horizon supervised objective with a physical-consistency penalty,

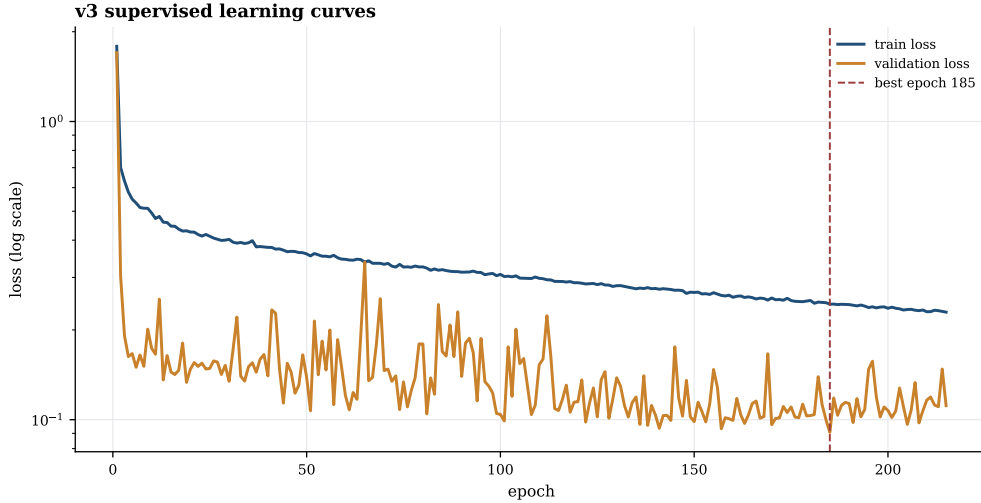
$$\mathcal{L}_{v3} = \sum_{k \in \{1,2,4\}} \frac{1}{N} \sum_i (\hat{T}_i(t+k) - T_i(t+k))^2 + \lambda_{\text{phys}} \text{ReLU}(|\Delta\hat{T}| - \Delta T_{\text{max}}), \quad (4)$$

where the multi-step term (horizons of 1, 2, and 4 transitions) enforces short-horizon rollout consistency and the penalty discourages physically implausible single-step jumps. The update $\hat{T}_{t+1} = T_t + \Delta\hat{T}_{t+1}$ carries *no* explicit time step: the learned increment is a per-transition quantity, the design choice revisited in Section 4.6.

Table 5: v3 configuration from the active model class and Hou-and-Evins reporting artifacts.

Item	Value
Input dimension	8
Temperature head	HeatFlowNetV2, 7,105 params
Power head	PowerNetV2, 1,377 params
Total parameters	8,482
Epoch budget	500
Batch size	256
Learning rate	1e-3
Optimizer	AdamW
Early stopping	patience=30; improvement_threshold=1e-6

The canonical v3 checkpoint is the early-stopped optimum of a 215-epoch supervised run: the best validation epoch is 185, at which the held-out one-step temperature fit reaches $R^2 = 0.991$. This single-step score is deliberately optimistic relative to the multi-step rollout error in Section 4.4, which is the metric that actually governs whether the surrogate can substitute for BOPTEST.

**Figure 3:** v3 supervised train/validation learning curves. Early stopping at the best validation epoch precedes any validation-loss upturn, indicating no overfitting at the selected checkpoint.

The v3 input vector is intentionally low-dimensional. It keeps only the thermal state, outdoor disturbance, cyclic time embedding, and direct- T_{sup} action coordinates required by the control problem. The action coordinate $a_0 \in [-1, 1]$ is mapped to an 18–35 °C supply-temperature command, and the power head is constrained through a non-negative output transform. These choices make v3 a control-oriented surrogate rather than a generic black-box forecaster.

Table 6: Feature scaling and physical constraints used by the Block 1 surrogate interface.

Variable	Scaling method	Parameters	Engineering role
surrogate_t_zone	affine min-max to [-1, 1]	min=15 C; max=35 C	Keeps the learned thermal state within the training support used by the direct-TSup surrogate.
surrogate_t_amb	affine min-max to [-1, 1]	min=-10 C; max=40 C	Normalizes outdoor-air temperature for the heat and power heads.
hour	cyclic sine/cosine	$\sin(2\pi \cdot \text{hour}/24)$; $\cos(2\pi \cdot \text{hour}/24)$	Avoids discontinuity between hour 23 and hour 0.
day	cyclic sine/cosine	$\sin(2\pi \cdot \text{day}/365)$; $\cos(2\pi \cdot \text{day}/365)$	Represents seasonal phase without an artificial year-end jump.
a0_raw	native action domain	[-1, 1] maps to 18..35 C supply temperature	Preserves the PPO action coordinate while keeping physical supply-temperature bounds explicit.
a1_raw	native action domain	[-1, 1] maps to 0..1 fan command	Keeps fan modulation aligned with the controller action range.
power_head_output	Softplus multiplied by P_MAX	P_MAX=5500 W	Guarantees non-negative predicted HVAC power.

4.3 Physics-Informed Twin and Inverse Calibration (v3.5)

Unlike the purely black-box v3 head, the v3.5 surrogate is built on a first-principles lumped-capacity zone balance with a single physical parameter, the zone thermal capacitance C_{zon} ,

$$C_{\text{zon}} \frac{dT_{\text{zone}}}{dt} = \dot{Q}(x; \phi), \quad \dot{Q}(x; \phi) = q_{\text{scale}} q_{\phi}(x), \quad (5)$$

where $\dot{Q}(x; \phi)$ is a learned net heat-flow head ($q_{\text{scale}} = 3000$ W). HVAC input, envelope coupling, and internal gains are absorbed into $\dot{Q}$ through the feature vector x ; there is no separately parameterized $1/R$ term. With control step Δt , (5) is integrated by an explicit Euler step,

$$\hat{T}_{t+1} = \text{clip}\left(T_t + \Delta t \frac{\dot{Q}(x; \phi)}{C_{\text{zon}}}, T_{\min}, T_{\max}\right). \quad (6)$$

To keep the inverse problem well posed, C_{zon} is not a free scalar but a positively reparameterized one,

$$C_{\text{zon}} = c_s (\text{softplus}(\rho) + c_{\min}/c_s), \quad C_{\text{zon}} \geq c_{\min} = 5 \times 10^4 \text{ J/K}, \quad (7)$$

with ρ the learnable log-parameter and $c_s = 10^5$ a fixed scale; this guarantees a physically positive capacitance throughout optimization. Because C_{zon} is a property of the building rather than of the data-generating policy, it is expected to remain meaningful across testcases, which is what makes the inverse identification below worthwhile.

The roadmap builds the canonical v3.5 artifact by two sequential preset runs. Stage A aligns telemetry and removes latency/bias artifacts. Stage B identifies C_{zon} on excitation-rich windows. Stage C then calibrates residual heads while preserving the physical parameter. The first preset, **block1_15min_episodeaware**, performs the temperature-side inverse identification: Stage B runs for 120 epochs and Stage C runs for 60 epochs. The second preset, **block1_15min_power_head_only**, loads the first summary JSON, freezes C_{zon} , skips Stage B, and refines only the power residual head for 80 epochs. This two-pass structure is why the power-head JSON reports **stage_b_epochs_ran=0**; it is not a missing calibration step.

Stage A estimates a one-step latency compensation, a temperature bias correction of -0.105 °C, and a postprocessed temperature RMSE of 0.238 °C. Stage B uses the top-excitation subset: 403 of 8058 training rows at excitation quantile 0.95. This filtering is important because C_{zon} is identifiable mainly during transients, not quasi-steady operation.

Table 7: Stage A telemetry-alignment operations from the calibration-summary JSON.

Operation	Estimated value	Effect
Latency compensation	1 step (15 min)	latency-search RMSE 0.378 C
Temperature bias removal	-0.1049 C	median residual offset
Power affine normalization	scale -0.702, bias 1148 W	global power-channel rescale
Post-Stage-A temperature RMSE	0.2383 C	baseline for Stage B
Post-Stage-A power MAE	514.4 W	baseline for Stage C

Table 8: Stage B excitation filtering and physical identification of C_{zon} .

Parameter	Value	Notes
C_{zon} prior	4.200×10^5 J/K	air volume x specific heat
C_{zon} after Stage B	4.413×10^5 J/K	+5.1% vs prior
Stage B epochs	120	episode-aware run
C_{zon} learning rate	10^{-3}	dedicated scalar LR
Excitation quantile	0.95	$ dT $ threshold 0.1759
Excitation rows	403 / 8058	mean score 0.240 vs 0.069

Formally, Stage B is a maximum-a-posteriori estimate of the single physical scalar,

$$\hat{C}_{\text{zon}} = \arg \min_C \sum_{i \in \mathcal{E}} (\hat{T}_i(C) - T_i)^2 + \frac{1}{2\sigma_p^2} (C - C^{\text{prior}})^2, \quad (8)$$

where $\mathcal{E}$ is the excitation subset and the quadratic term is a weak Gaussian prior centred on C^{prior} . The excitation filter is justified by identifiability: in quasi-steady operation the sensitivity $\partial \hat{T} / \partial C \rightarrow 0$, so steady windows carry almost no information about C_{zon} and would merely pull the estimate back to the prior. Restricting $\mathcal{E}$ to the top-excitation transients (403 of 8058 rows) concentrates the identification where $\partial \hat{T} / \partial C$ is large, which is why the mean excitation score on $\mathcal{E}$ is several times the corpus mean (Table 8).

The intermediate Stage B estimate is $C_{\text{zon}} = 4.413 \times 10^5$ J/K, and the relative update over the physical prior is

$$\eta_C = \frac{C_{\text{zon}}^{\text{final}} - C_{\text{zon}}^{\text{prior}}}{C_{\text{zon}}^{\text{prior}}} = 0.0506, \quad (9)$$

where $C_{\text{zon}}^{\text{final}} = 4.413 \times 10^5$ J/K. The update remains inside the prior physical plausibility band.

Beyond the point estimate, two complementary uncertainty checks apply. The optimizer is well conditioned (over the final 20 Stage B epochs the C_{zon} trajectory is stable to within $\pm 1,005$ J/K, 0.23%). More importantly, a Laplace/Fisher identifiability interval — obtained from the local curvature of the one-step temperature SSE with respect to C_{zon} (residual heads fixed) over the high-excitation transients ($n = 1838$) — gives a 1σ uncertainty of 6.6% and a 95% interval $[3.845, 4.980] \times 10^5$ J/K. The +5.1% data-driven update therefore sits within roughly one standard error of the prior: C_{zon} is identifiable with a moderate, honestly bounded signal rather than a tightly pinned value. Physically, the identified capacitance corresponds to an equivalent thermal mass of $C_{\text{zon}}/c_{p,\text{air}} \approx 439$ kg of air ($c_{p,\text{air}} = 1005 \text{ J kg}^{-1} \text{ K}^{-1}$), or roughly 366 m^3 of equivalent air volume at $\rho = 1.2 \text{ kg m}^{-3}$. This is the expected order of magnitude for the `bestest_air` zone augmented by the effective thermal mass of its internal surfaces, which is exactly why the data-driven update over the air-only prior is small (+5%) rather than large.

Physical-consistency audit. As a final check we probe the learned dynamics directly: sweeping the supply-temperature command across its range with the state held fixed, the predicted next-step zone temperature increases with T_{sup} in 100% of 400 sampled states (correct sign in every case), with a mean sensitivity of $+0.061$ °C per °C and a mean Spearman rank correlation of 0.91 (Figure 4). The response is monotone in the physically expected direction; the small departures from strict monotonicity reflect the neural residual head and stay well inside the surrogate temperature tolerance.

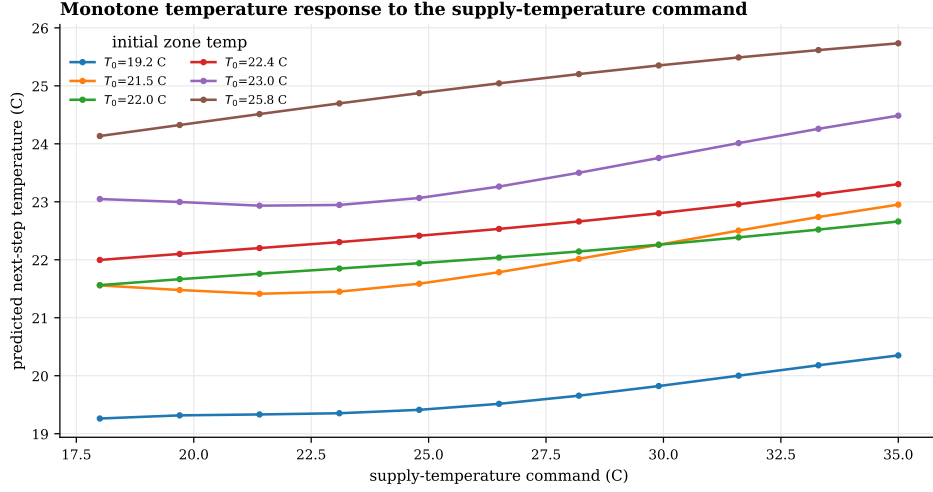


Figure 4: Physical-consistency audit: predicted next-step zone temperature versus the supply-temperature command for representative initial states. The response is monotone increasing, so the learned dynamics respect the correct sign of the control.

Stage A/B/C calibration improves prediction while keeping the physical parameter bounded

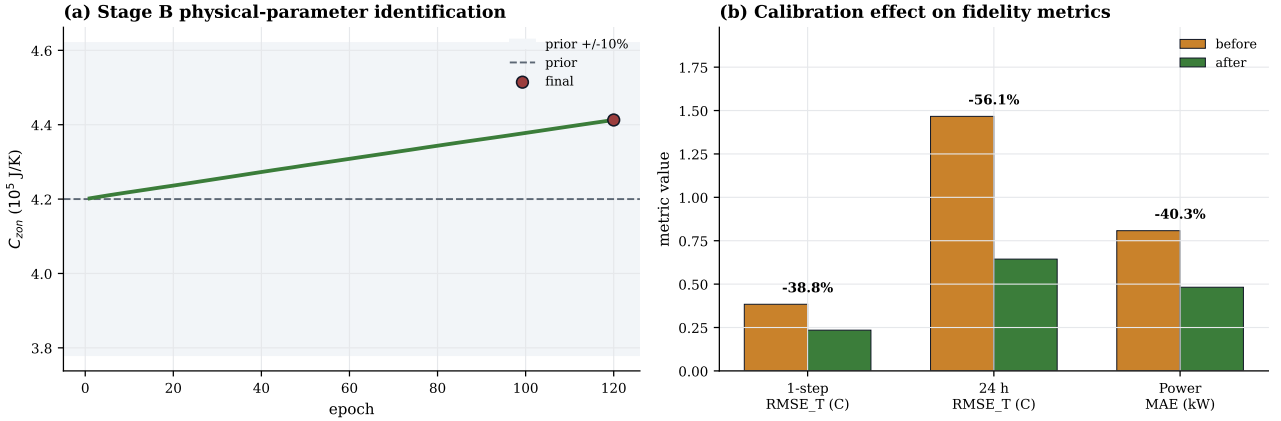


Figure 5: Stage A/B/C calibration diagnostics. Panel (a) shows physical-parameter convergence inside the prior band; panel (b) shows the measured improvement in temperature and power fidelity.

Table 9: Stage A/B/C calibration summary from the canonical JSON and rollout artifacts.

Metric	Before	After	Relative change
1-step RMSE_T ($^{\circ}\text{C}$)	0.384	0.235	-38.8%
24 h rollout RMSE_T ($^{\circ}\text{C}$)	1.466	0.644	-56.1%
Power MAE (W)	808	482	-40.3%
C_{zon} ($\times 10^5$ J/K)	4.200	4.413	+5.1%

Table 10: Key design choices for the v3/v3.5 surrogate family and their engineering rationale. A formal sensitivity sweep over these settings is identified as future work in Section 4.7.

Choice	Value	Rationale
Hidden dimension	64	smooth, low-curvature response surface for stable policy gradients
Optimizer / LR	AdamW / 1e-3	decoupled weight decay; cosine annealing
Batch size	256	variance/throughput trade-off on the prepared corpus
Heat-flow scale q_{scale}	3000 W	keeps $q_{\phi}(x)$ near unit range for conditioning
Capacitance floor c_{min}	5×10^4 J/K	physical lower bound enforced by reparameterization
Excitation quantile	0.95	isolates transient windows where $\partial \hat{T} / \partial C$ is large
C_{zon} scalar LR	10^{-3}	dedicated rate for the single physical parameter

4.4 Multi-Step Validation and Matched-Corpus Ablation

4.4.1 Multi-step predictive validation

For horizon h , the rollout error and RMSE are

$$e_i(h) = \hat{T}_i(t+h) - T_i(t+h), \quad (10)$$

$$\text{RMSE}_T(h) = \sqrt{\frac{1}{N} \sum_{i=1}^N e_i(h)^2}. \quad (11)$$

Engineering tolerance is reported as an absolute-error CDF,

$$F_{|e|}(\epsilon) = \Pr(|\hat{T} - T_{\text{BOPTEST}}| \leq \epsilon). \quad (12)$$

The coefficient of determination follows the standard definition $R^2 = 1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$ against the held-out mean, so a *negative* 24 h R^2 (as for the hourly v3 in Table 11) means the long-horizon rollout is worse than predicting that mean. The error tail is summarized by the 95th-percentile absolute error $\text{P95} = \inf\{\epsilon : F_{|e|}(\epsilon) \geq 0.95\}$.

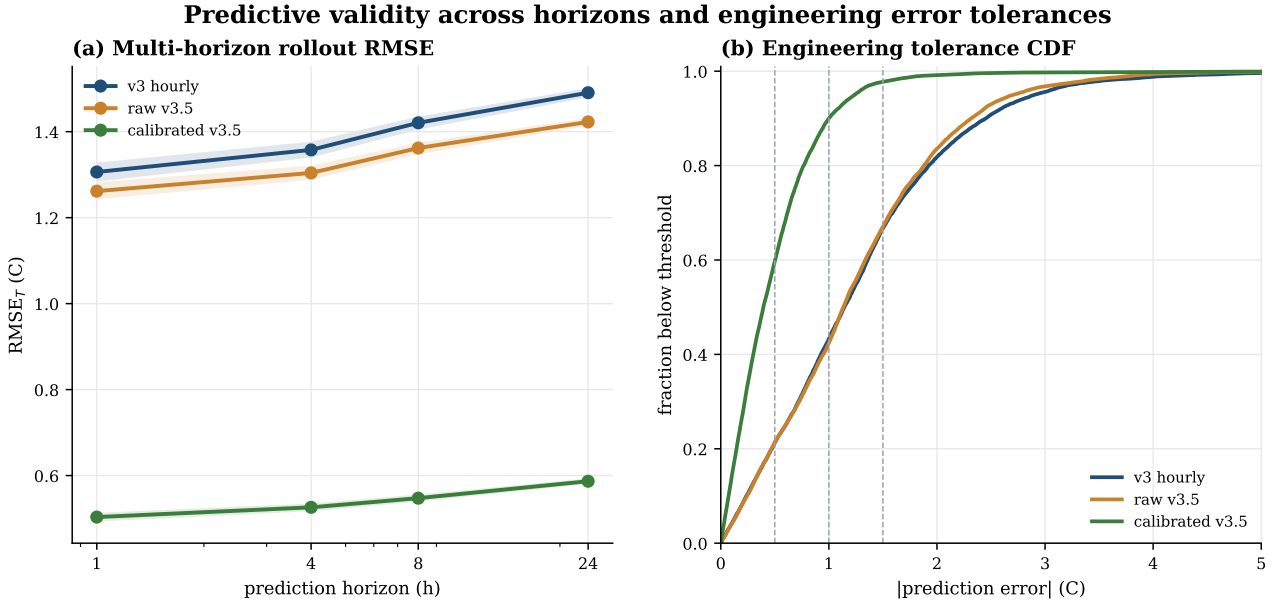


Figure 6: Predictive-validity diagnostics from prepared BOPTEST rollouts. Confidence bands are bootstrapped from real window-level errors, and the CDF reports engineering error tolerances at 0.5, 1.0, and 1.5 °C.

The calibrated v3.5 surrogate reaches a 24 h rollout $\text{RMSE}_T = 0.644$ °C, compared with raw v3.5 at 1.466 °C and the canonical v3 hourly reference at 1.557 °C.

Table 11: Multi-horizon teacher-forced rollout fidelity from the Hou-and-Evins predictive-validity table. Cells are temperature RMSE ($^{\circ}\text{C}$) at each horizon; the final column is the 24 h temperature R^2 (n/a where not computed for the auxiliary baselines).

Model	1 h	4 h	8 h	24 h	24 h R^2_T
v3 (hourly)	1.471	1.579	1.579	1.558	-1.411
v3 (15-min matched)	0.875	0.874	0.873	0.876	n/a
raw v3.5	1.491	1.491	1.491	1.466	n/a
calibrated v3.5	0.321	0.575	0.638	0.644	+0.588

Table 11 exposes the qualitative split between the two surrogate families. The hourly v3 model has a *negative* 24 h temperature R^2 – it is worse than predicting the historical mean over long horizons – yet it remains a usable control surrogate because PPO needs smooth local gradients, not long-horizon physical realism. The calibrated v3.5 surrogate instead keeps a positive 24 h R^2 and bounds its error tail: its 24 h P95 absolute error is 1.207°C , so even in worst-case windows the calibrated twin stays within that band 95% of the time. Reporting raw v3.5 and the corpus-matched v3 in the same table lets the matched-corpus attribution in Section 4.4.2 be read directly against the per-horizon numbers. As a model-free reference, a naive persistence forecast ($\hat{T}(t+h) = T_t$) on the same episodes gives 0.397°C at 1 h and 0.969°C at 24 h: calibrated v3.5 beats persistence at every horizon, whereas the hourly v3 is actually *worse* than persistence at 24 h (1.557 vs 0.969°C), which quantitatively confirms that v3 is a control surrogate rather than a forecaster.

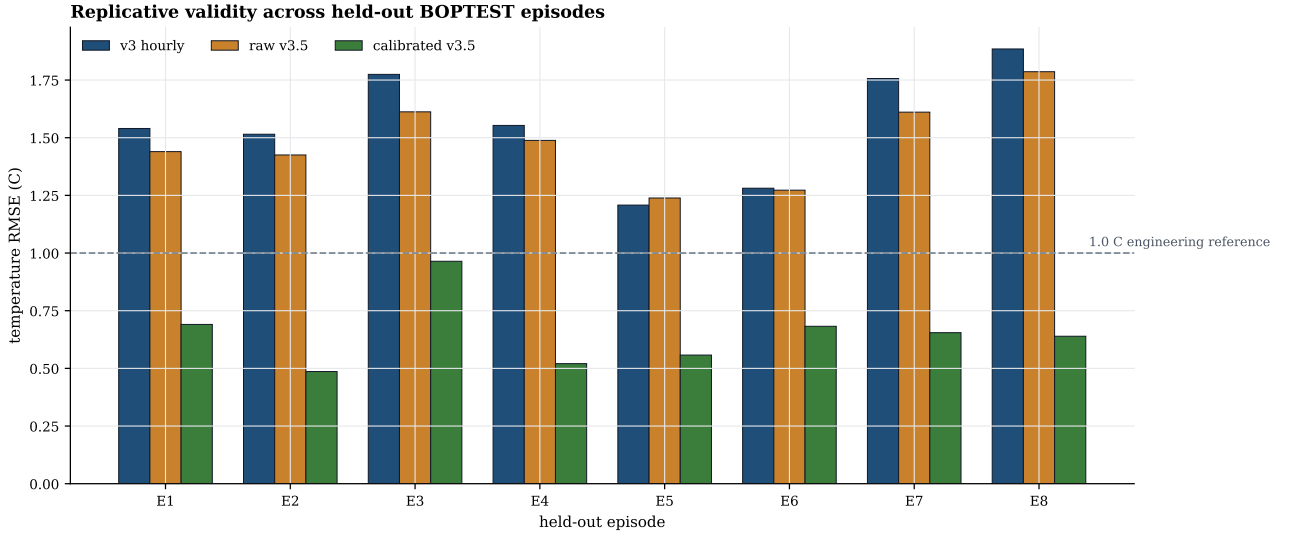


Figure 7: Replicative validity across the eight held-out BOPTEST episodes used by the Block 1 prepared-rollout artifacts. The dashed line marks a 1°C engineering reference.

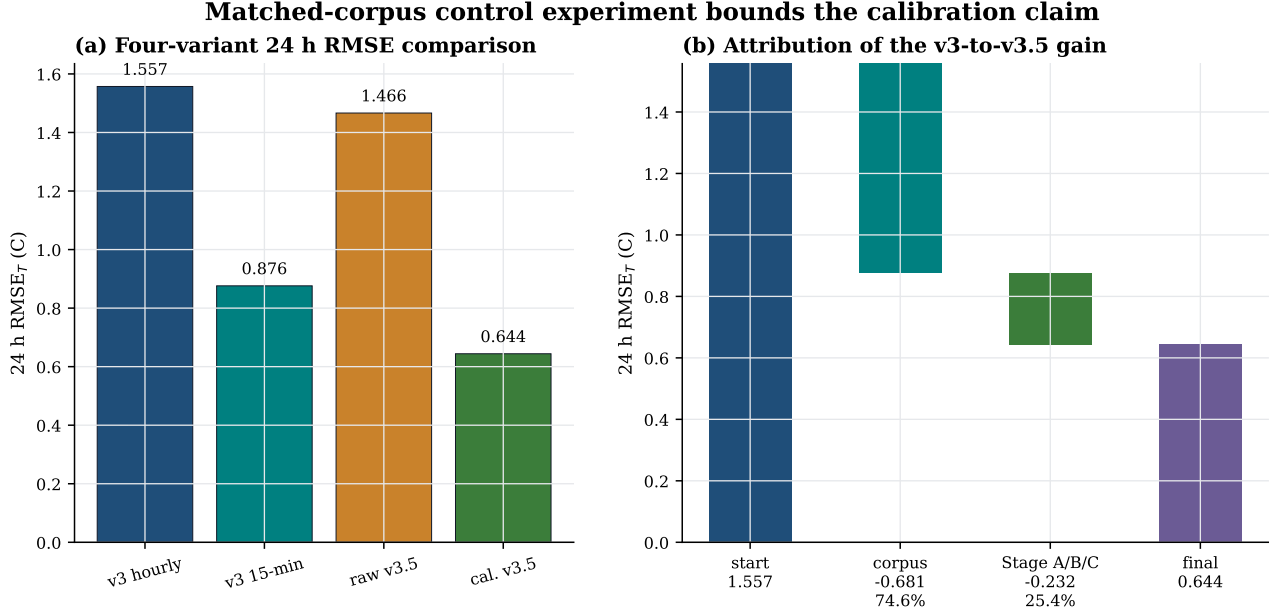
The episode-level diagnostic is important because a low aggregate RMSE can hide a single unstable rollout. The calibrated v3.5 artifact remains below the 1°C reference in all eight held-out episodes; its best episode is peak heat thermostatic at 0.486°C and its worst episode is typical heat hdrl at 0.964°C . This supports predictive validity as a repeated-episode result rather than a single-window artifact.

4.4.2 Matched-corpus ablation

The roadmap includes a corpus-matched v3 retraining step because the original v3 and canonical v3.5 artifacts use different data resolutions. This is not a cosmetic check: without it, a reviewer could correctly attribute the v3.5 gain to the 15-minute corpus rather than Stage A/B/C calibration.

Table 12: Corpus-matched comparison.

Variant	Step (s)	Calibration	24 h RMSE _T
v3 hourly	3600	n/a (control-oriented black box)	1.557
v3 15-min matched	900	n/a (control-oriented black box)	0.876
raw v3.5	900	OFF (no Stage A/B/C)	1.466
calibrated v3.5	900	FULL (Stage A + B + C, C _{zon} = 4.413e5 J/K)	0.644

**Figure 8:** Matched-corpus attribution. The v3-to-v3.5 gain is split into the 15-minute corpus contribution and the Stage A/B/C calibration contribution.

The attribution is an additive split of the legacy-v3-to-calibrated-v3.5 gap,

$$\underbrace{\Delta_{\text{total}}}_{\text{v3 hourly} \rightarrow \text{cal. v3.5}} = \underbrace{\Delta_{\text{corpus}}}_{\text{v3 hourly} \rightarrow \text{v3 15-min}} + \underbrace{\Delta_{\text{cal}}}_{\text{v3 15-min} \rightarrow \text{cal. v3.5}}, \quad (13)$$

which holds exactly because the matched-architecture path is constructed to be additive (the intermediate point is the same v3 architecture retrained on the 15-min corpus). The total 24 h RMSE drop from the legacy v3 reference to calibrated v3.5 is 0.913 °C. The matched-corpus report attributes 0.681 °C (74.6%) to the corpus shift and 0.232 °C (25.4%) to Stage A/B/C. A second, equally valid attribution path runs through the raw v3.5 backbone: corpus and architecture together explain only 9.9% of the drop (0.091 °C), while Stage A/B/C applied to the physical backbone explains 90.1% (0.822 °C). The two paths differ because corpus and architecture effects interact non-additively; the matched-architecture split is quoted as the primary attribution because both of its endpoints are control-architecture surrogates. Either way the calibration claim is positive but bounded: corpus resolution and physical calibration jointly produce the final fidelity improvement.

4.5 Runtime Feasibility

The final Block 1 roadmap command builds the speed benchmark. The benchmark compares the same BOPTEST RTE HTTP loop used by the paper against local surrogate stepping under a 900 s control step. The result is not a controller claim; it only establishes feasibility for downstream policy optimization.

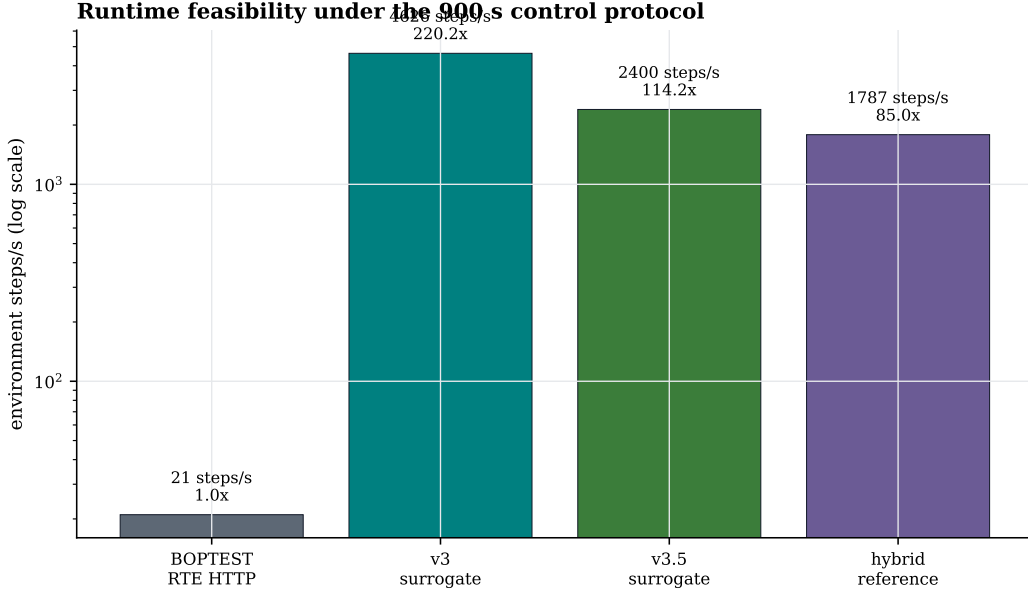


Figure 9: Runtime feasibility benchmark. The hybrid reference is included only to demonstrate downstream feasibility; controller utility is evaluated in Block 2.

Table 13: Backend throughput under the 900 s control protocol from the speed-benchmark artifact (single CPU thread, BOPTTEST reached through the same HTTP-Docker interface used in training).

Backend	Steps/s	Median (ms)	P95 (ms)	Speed-up
BOPTTEST RTE HTTP loop	21.0	14.915	19.251	1.0×
v3 surrogate	4,626.3	0.165	0.273	220.2×
v3.5 calibrated	2,399.5	0.359	0.495	114.2×
hybrid v3+v3.5	1,786.8	0.507	0.648	85.0×

At the hybrid throughput, a single 5×10^6 -step policy-optimization run costs about 46.6 minutes of pure environment time, versus roughly 66.1 hours on the live BOPTTEST RTE HTTP loop. The calibrated v3.5 surrogate is 114.2x faster than the BOPTTEST RTE HTTP loop, and the hybrid reference remains 85.0x faster. This closes the runtime part of Block 1: the surrogate family is not only more accurate after calibration, but also fast enough to support repeated policy-training experiments without using the RTE HTTP loop as the inner training environment.

4.6 Step-Size Design Disclosure

The canonical v3 checkpoint is trained at a 3600 s step, whereas all downstream policy optimization runs at the 900 s control step. Because the v3 transition is $\hat{T}_{t+1} = T_t + \Delta \hat{T}_{t+1}$ with no explicit timestep scaling, the hourly-trained increment is applied unchanged at the 900 s step. This is the structural reason the legacy v3 carries a 1.557 °C 24 h rollout error, while the corpus-matched v3 trained directly on 900 s transitions reaches 0.876 °C (Table 12). The mismatch is preserved deliberately and is auditable: every controller-facing KPI in the later blocks is measured on the live BOPTTEST RTE simulator, not on the surrogate, so the surrogate’s only role is to supply a smooth gradient signal during policy updates. Re-running the pre-registered downstream experiments with the matched-corpus checkpoint would break the frozen audit chain, so the matched-corpus model is used only for the predictive-fidelity attribution of Section 4.4.2. By contrast with v3, the v3.5 forward pass (6) carries an explicit Δt ; a warm-started v3.5 therefore rescales the inherited v3 heat term by

$$s = s_{\text{legacy}} \frac{C_{\text{init}}}{\Delta t_{\text{legacy}} q_{\text{scale}}}, \quad (14)$$

so that the per-step increment is dimensionally consistent at the 900 s runtime step. This time-scaling is exactly what the v3 update lacks.

4.7 Limitations

Validation scope. The digital-twin validation in this section is scoped to the *temperature* channel, which is the comfort-relevant quantity and is the channel that meets engineering tolerance (24 h rollout RMSE 0.644 °C, P95 1.207 °C, near-zero bias NMBE -0.117 °C on a mean zone temperature of 22.3 °C). The instantaneous *power* channel is reported but *not* claimed as fully calibrated. On the canonical (power-head) checkpoint its mean absolute error is 482 W on a mean HVAC power of 1002 W, giving a coefficient of variation of the RMSE of about 69% and a normalized mean bias error of -12.4%. The CV(RMSE) exceeds the ASHRAE Guideline 14 threshold of 30% and the NMBE is just beyond the $\pm 10\%$ band, but both are expected for an *instantaneous* 15-min power signal of a cycling HVAC system: Guideline 14 is formulated for energy aggregated to hourly or monthly resolution, not for instantaneous power. Because every downstream controller-facing KPI is evaluated on the live BOPTEST simulator rather than on the surrogate power channel, this residual power error does not propagate into the control results; it is disclosed here as a genuine limitation of the surrogate power head and as one motivation for the power-disagreement regularizer introduced later.

Other limitations. (i) A single testcase (`bestest_air`) is used; cross-building generalization is deferred to the Block 3 transferability study. (ii) The supervised held-out split is contiguous (autumn-only), so the one-step R^2 is optimistic and all reported control claims rely on external BOPTEST windows rather than this split. (iii) Each surrogate is trained from a single seed; a multi-seed variance study and a formal sensitivity sweep over the design choices of Table 10 are future work. (iv) The direct- T_{sup} interface abstracts the actuator chain beyond the Stage A latency term, and the hourly-vs-15-min step mismatch is preserved by design (Section 4.6).

4.8 Block 1 Conclusion

Block 1 establishes four results that are carried forward into the rest of the article. First, the v3 direct- T_{sup} surrogate is a deliberately compact 8D dual-head model with 8,482 trainable parameters, trained on 51,200 hourly transitions; it is fast and smooth, but its 24 h rollout error remains 1.557 °C. Second, the v3.5 Stage A/B/C inverse-calibration path identifies a physically bounded zone capacitance of 4.413×10^5 J/K and reduces the prepared-corpus 24 h rollout error to 0.644 °C. Third, the corpus-matched control experiment prevents an overclaim: 74.6% of the v3-to-v3.5 fidelity gain is explained by the shift to the 15-minute corpus, while 25.4% is attributable to Stage A/B/C calibration. Fourth, the speed benchmark shows that the calibrated v3.5 model remains 114.2x faster than BOPTEST RTE HTTP. Therefore Block 1 supports the article’s digital-twin claim, but it does not yet support a controller-transfer claim; that question is tested separately in Results II under live BOPTEST validation.